Advanced Macroeconomics (cod. 20285)
Problem set 4

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- Due date: May 3rd, by **23:59** CEST;
- Write your solutions with a word processor (LaTeX would be preferred);
- **Grading** is based on **either Exercise 1 or 2** (randomly chosen) and **Exercise 3**;
- **Optional** questions will **not be graded**. They serve as additional practice exercises;
- Please submit your solutions to henri.graul@unibocconi.it;
- Submit only **one** pdf for each group, don't forget to specify the names of the members of the group on the first page. Please also **indicate your names in the document title**, i.e. for example "PS4Graul/Trigari.pdf".

Exercise 1: Optimal price setting in the Taylor model

In this exercise, we explore the Taylor model of staggered price setting, where each firm resets its price for N periods. In each period, a fraction $\frac{1}{N}$ of firms reset their prices, while the remaining firms keep their prices fixed. Each firm i produces a differentiated good with technology:

$$Y_t(i) = A_t N_t(i),$$

where A_t represents the level of technology, evolving exogenously as $a_t = \log A_t$ with a stationary stochastic process. A firm resetting its price P_t^* in period t seeks to maximize its expected profits over N periods:

$$\max_{P_t^*} \sum_{k=0}^{N-1} \mathbb{E}_t \left\{ Q_{t,t+k} \left(\frac{1}{P_{t+k}} \right) \left(P_t^* Y_{t+k|t} - \Psi_{t+k}(Y_{t+k|t}) \right) \right\}$$

subject to:

$$Y_{t+k|t} = \left(\frac{P_t^*}{P_{t+k}} \right)^{-\epsilon} C_{t+k},$$

where $Q_{t,t+k} = \beta^k \left(\frac{C_{t+k}}{C_t} \right)^{-\sigma}$ is the stochastic discount factor, $\Psi_{t+k}(Y_{t+k|t})$ are the nominal cost, and C_{t+k} is aggregate consumption in period $t+k$.

1. Show that the optimal reset price P_t^* must satisfy the first-order condition:

$$\sum_{k=0}^{N-1} \mathbb{E}_t \left\{ \left(\frac{1}{P_{t+k}} \right) Q_{t,t+k} Y_{t+k|t} \left(P_t^* - \mathcal{M}\psi_{t+k} \right) \right\} = 0,$$

where $\psi_{t+k} = \Psi'(\cdot)$ is the nominal marginal cost and $\mathcal{M} = \frac{\epsilon}{\epsilon-1}$.

[Hint: Set up the first-order condition by differentiating the profit function with respect to P_t^*. Remember to substitute the demand function and use the stochastic discount factor for nominal payoffs.]

2. Derive the log-linearized version of the optimal price-setting rule around a zero-inflation steady state:

$$p_t^* = \mu + \sum_{k=0}^{N-1} \omega_k \mathbb{E}_t \left\{ \varphi_{t+k} \right\},$$

where $\omega_k = \frac{\beta^k(1-\beta)}{1-\beta^N}$, $\varphi_{t+k} = \log \psi_{t+k}$, and $\mu = \log \mathcal{M}$.

Interpret this condition.

[Hint: Approximate the optimal price around a zero-inflation steady state by log-linearizing the price-setting rule. This will involve taking logarithms and expectations.]

3. How does the Taylor model's price-setting rule differ from the one generated by the Calvo model? Discuss the economic intuition behind these differences, focusing on the implications for inflation persistence. How do they compare in terms of tractability?

4. Recalling that the aggregate price index P_t is defined by:

$$P_t = \left(\int_0^1 P_t(i)^{1-\epsilon} \, di \right)^{\frac{1}{1-\epsilon}},$$

show that, around a zero-inflation steady state, the (log) price level will satisfy:

$$p_t = \frac{1}{N} \sum_{k=0}^{N-1} p_{t-k}^*.$$

[Hint: Argue why we have symmetric pricing for firms within each cohort and derive the relationship between the log price level p_t and the optimal reset prices p_{t-k}^*.]

Exercise 2: The Phillips Curve with Price Indexation

In the New Keynesian model, there is a probability θ that a firm will not be able to change its nominal price in a given time period. Firms that do have an opportunity to change price at time t choose a reset price p_t^* according to:

$$p_t^* = (1 - \beta\theta) \sum_{\ell=0}^{\infty} (\beta\theta)^\ell \mathbb{E}_t [p_{t+\ell} + \gamma \widehat{mc}_{t+\ell}],$$

where β is the discount factor, p_t is the price level, $\widehat{mc}_t$ is average marginal cost, and $\gamma = \frac{1-\alpha}{1-\alpha+\alpha\epsilon}$ is a positive coefficient. All variables are given as log deviations from their steady-state values. The general price level p_t is determined by:

$$p_t = (1 - \theta) \sum_{\ell=0}^{\infty} \theta^\ell p_{t-\ell}^*.$$

1. Provide an intuitive explanation of the two equations above.

2. Show that the reset-price and price-level equations are equivalent to:

$$p_t^* = \beta\theta\mathbb{E}_t p_{t+1}^* + (1-\beta\theta)(p_t + \gamma\widehat{mc}_t),$$

   and

$$p_t = \theta p_{t-1} + (1-\theta)p_t^*.$$

3. By combining the two equations, derive the New Keynesian Phillips curve for inflation $\pi_t = p_t - p_{t-1}$:

$$\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa\widetilde{y}_t,$$

   using that $\widehat{mc}_t = \left(\sigma + \frac{\alpha+\phi}{1-\alpha}\right)\widetilde{y}_t$, and where $\kappa = \gamma\left(\sigma + \frac{\alpha+\phi}{1-\alpha}\right)\left(\frac{(1-\theta)(1-\beta\theta)}{\theta}\right)$

Next suppose that when a firm does not have an opportunity to choose a new reset price optimally at time t (probability θ), the firm adjusts its old nominal price in line with any past inflation π_{t-1} that has occurred.[1] This means the equation for the price level is now:

$$p_t = \theta(p_{t-1} + \pi_{t-1}) + (1-\theta)p_t^*.$$

When choosing the reset price p_t^*, firms take account of the indexation of prices to past inflation. Hence the new equation for the reset price is:

$$p_t^* = (1-\beta\theta)\sum_{\ell=0}^{\infty}(\beta\theta)^\ell\mathbb{E}_t[p_{t+\ell} - (p_{t+\ell-1} - p_{t-1}) + \gamma\widehat{mc}_{t+\ell}].$$

4. Explain the intuition for the new reset-price equation.

5. Derive the new Phillips curve. Comment on any differences relative to the standard New Keynesian Phillips curve.

   [Hint: Start by putting the reset-price equation in a simpler recursive form.]

Exercise 3: The Phillips Curve

Part A: The Classic Phillips Curve

For this exercise, download the following series at quarterly frequency from 1960:Q1 through 2025:Q4 from FRED:

- u_t = *Unemployment Rate* (Unit: Percent, Seasonally Adjusted);
- π_t = *Personal Consumption Expenditures: Chain-type Price Index* (Unit: Percent Change from Year Ago, Seasonally Adjusted);

1. Construct a series for expected price inflation, measured as $\widetilde{\mathbb{E}}_t[\pi_{t+1}] = \frac{1}{4}(\pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4})$. Briefly interpret and comment on this measure of inflation expectations.

2. Create a scatterplot of u_t (x-axis) and $(\pi_t - \widetilde{\mathbb{E}}_t[\pi_{t+1}])$ (y-axis) for the entire period. Fit a simple OLS regression line and plot it in the same graph. Briefly describe your findings. What is $(\pi_t - \widetilde{\mathbb{E}}_t[\pi_{t+1}])$ measuring?

[1] For a more detailed treatment, see for example "Trend Inflation, Indexation, and Inflation Persistence in the New Keynesian Phillips Curve" by Cogley and Sbordone (AER 2008).

3. Repeat the exercise from above for the following subperiod:

   a. 1960:Q1 – 1986:Q4 (blue)
   b. 1987:Q1 – 2007:Q4 (green)
   c. 2008:Q1 – 2019:Q4 (red)
   d. 2020:Q1 – 2025:Q4 (black)

   Compare the scatterplots and slopes over time and discuss.

Part B: Household expectations

4. Read and summarize the key question and findings of "Is the Phillips Curve Alive and Well after All? Inflation Expectations and the Missing Disinflation" by Coibion and Gorodnichenko (AEJ: Macro 2015).

In this exercise, we focus on the third potential explanation for the missing disinflation discussed in Section IV of the paper. Begin by downloading the following time series:

- $\mathbb{E}_t^{MSC}[\pi_{t+4}]$ = *Household inflation forecast, Median 1-year ahead* (Source: University of Michigan - Survey of Consumers, Table 32, time span: 1978:Q1-2025:Q4).

Next, we will estimate the Phillips curve using household forecasts. Specifically, we consider a Friedman (1968)-type expectations-augmented Phillips curve:

$$\pi_t - \mathbb{E}_t[\pi_{t+1}] = c + \kappa u_t + v_t, \tag{1}$$

as in equation (1) of Coibion and Gorodnichenko (2015). We are interested in estimating κ, the slope of the Phillips curve. To do so, we will implement different measures of inflation expectations, $\mathbb{E}_t[\pi_{t+1}]$.

5. First, plot, describe and compare the time series of the two measures of inflation expectations, $\widetilde{\mathbb{E}}_t[\pi_{t+1}]$ and $\mathbb{E}_t^{MSC}[\pi_{t+4}]$). Why is it useful to use the median (instead of the mean for example) for household inflation expectations?

6. Next, estimate (1) assuming that $\mathbb{E}_t[\pi_{t+1}] = \mathbb{E}_t^{MSC}[\pi_{t+4}]$. Create a scatter plot and include your estimated regression line, as in Figure 6 Panel B of the paper. Relate your findings to the results of Coibion and Gorodnichenko (2015) and discuss.

7. Briefly discuss the OLS estimation of the slope of the Phillips curves in 3.2, 3.3. and 3.6.

   [Hint: Specifically, consider the identification problem of simultaneity bias in the given case. You may want to have a look at Section 2 and, more importantly, Section 3 of McLeay and Tenreyro (2019).]

Part C: Non-linear Phillips curves

8. Read "The Slanted-L Phillips Curve" by Benigno and Eggertsson (2024) (AEA Papers and Proceedings, 2024). Briefly summarize the authors' key arguments.

9. Replicate Figure (1) of the paper for the US. To do so, simply perform the following steps:

(a) Download quarterly data from the following series from Q1 2000 through Q4 2025 from FRED:

- V_t = *Job Openings: Total Non-Farm* (Unit: Level in Thousands, Seasonally Adjusted);
- U_t = *Unemployment Level* (Unit: Thousands of Persons, Seasonally Adjusted);

(b) For each period, compute the V_t/U_t ratio. Finally, create a scatter plot of V_t/U_t (x-axis) and π_t (y-axis).

(c) Briefly describe your findings and discuss.[2]

10. Lastly, read Giulia Gitti's Job Market Paper (JMP, 2024). Explain her approach of addressing potential endogeneity when relating labor market tightness and inflation. How does this relate to your findings in 3.8 and 3.9 ?

[2]For a more detailed descriptive background see Benigno and Eggertson(2024) and Benigno and Eggertson(2024).

OPTIONAL EXERCISES

Note that the notation in these exercises might differ slightly from that used in the lectures.

Exercise 4: Optimal monetary policy under Price Indexation

In this exercise, we will analyse how partial price indexation by firms impacts optimal monetary policy design. In this case, the second-order approximation to the household's welfare losses takes the form:

$$\frac{1}{2}\mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t(\vartheta x_t^2+\{\pi_t-\zeta\pi_{t-1}\}^2)\right],$$

where ζ denotes the degree of price indexation to past inflation.
The equation describing the evolution of inflation is now given by:

$$\pi_t-\zeta\pi_{t-1}=\kappa x_t+\beta\mathbb{E}_t[\pi_{t+1}-\zeta\pi_t]+u_t,$$

where u_t represents an exogenous i.i.d cost-push shock.

1. Briefly explain the relationship between $x_t, \widetilde{y}_t$ and u_t. Why are we using x_t when studying optimal monetary policy? Next, compare the Phillips curve in this exercise with the one you obtained in Exercise 1.5. Under what condition are they congruent?

2. Determine the optimal policy under discretion. Specifically, we assume that for simplicity, the monetary authority seeks to minimize each period *only* the short-term losses, $\vartheta x_t^2+\{\pi_t-\zeta\pi_{t-1}\}^2$. This implies that the central bank does not set-up a dynamic problem, given that past inflation is a state variable in this setting. Hence it faces the following constrained optimization problem:

$$\min_{x_t,\pi_t}\vartheta x_t^2+\{\pi_t-\zeta\pi_{t-1}\}^2$$

s.t.

$$\text{s.t.}\quad\begin{cases}\pi_t-\zeta\pi_{t-1}=\kappa x_t+\beta\mathbb{E}_t[\pi_{t+1}-\zeta\pi_t]+u_t\\ \pi_{t-1}\text{ given}\end{cases}$$

3. Determine the optimal policy under commitment.

4. Discuss how the degree of indexation ζ affects the optimal responses to a transitory cost-push shock under the previous two scenarios.

5. *[Extra Optional:* Determine optimal policy under discretion *without* the simplifying assumption made in 2.2.]

Exercise 5: Government Spending Shocks and Optimal Monetary Policy

Consider an economy in which the government has a time-varying *exogenous* share of purchases in output γ_t:

$$G_t=\gamma_t Y_t,$$

where γ evolves exogenously, with $\gamma_t < 1$.
The economy's resource constraint reads:

$$Y_t = C_t + G_t.$$

The production function reads:

$$Y_t = A_t N_t,$$

where A_t is labor productivity, evolving exogenously as usual.
The representative household maximizes the intertemporal utility

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right\}$$

under a standard sequence of budget constraints. Assume $\sigma \geq 1$ and $\varphi \geq 0$.

1. Formulate the social planner's problem and derive the first-order conditions for efficiency. Derive an expression for the efficient level of output, denoted as Y_t^e. What are its determinants? In particular, what is the effect of a rise in γ_t?

2. Assume now that the economy is characterized by imperfect competition and price stickiness à la Calvo, and define ε as the elasticity of substitution between varieties. Write the expression for the marginal cost and use it, along with the price-setting condition from profit maximization and the labor supply condition from household optimization, to derive the expression for the natural level of output, Y_t^n. What are the determinants of Y_t^n? What is the effect of a rise in γ_t? Discuss.

3. Compare the expressions for the efficient level of output, Y_t^e, and the natural level of output, Y_t^n, by taking their ratio. Do they align? If not, explain why.

   [Hint: Your comparison should be based on two key variables: the markup and the government spending share.]

4. Assume now that in the steady state $Y^n = Y^e$. As shown from your answer to point 3, this requires $1 - \gamma = 1/\mathcal{M}$ or, equivalently, $\gamma = 1/\varepsilon$.

   Define two alternative measures of the output gap:

$$x_t \equiv \log Y_t - \log Y_t^e = y_t - y_t^e, \tag{2}$$

$$\tilde{y}_t \equiv \log Y_t - \log Y_t^e = y_t - y_t^n. \tag{3}$$

   Let a welfare-based loss function of the central bank be of the general form:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \pi_t^2 + \omega \left(output\ gap_t \right)^2 \right\}, \tag{4}$$

   subject to a Phillips curve constraint given by:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \left(output\ gap_t \right). \tag{5}$$

   Which measure of the output gap should appear in the Phillips curve, (2) or (3)? Which one should be used in the loss function? Provide an explanation.

5. Derive an expression for the gap, if any, between the efficient and natural levels of output, u_t, defined as $u_t \equiv \kappa(y_t^e - y_t^n)$. Determine whether a shock to government spending (to γ_t) can be interpreted as a cost-push shock.

6. Characterize the effects of a cost-push shock, u_t, under the optimal discretionary policy. Is there a stabilization bias? Is the policy time-consistent?